Twelve-critical graphs with $\left(\frac{2}{5}+o(1)\right)n^2$ edges

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Abstract

Let $f_k(n)$ denote the maximum number of edges in a k-critical graph on n vertices. We construct a sequence of twelve-critical graphs G_s with $|V(G_s)| \to \infty$ and

$$e(G_s) = \left(\frac{2}{5} + o(1)\right)|V(G_s)|^2.$$

Since $2/5 > 3/8$, this disproves $f_{12}(n) \sim 3n^2/8$, and hence the general asymptotic formula proposed in Erdős Problem 917. The construction combines a coloring module of Pegden with K_5-saturated graphs of sublinear maximum degree.

1 Introduction

All graphs are finite and simple. A graph is k-critical if it has chromatic number k and every proper subgraph is $(k-1)$-colorable; for graphs without isolated vertices this is equivalent to requiring only that deleting any edge lowers the chromatic number, the convention of [4]. Write $e(G) = |E(G)|$, and let $f_k(n)$ be the largest edge count on n vertices under that convention. Erdős [3, p. 28] conjectured that, for fixed $k \geq 6$ (in the formulation of [4]),

$$f_k(n) \sim \frac{1}{2}\left(1 - \frac{1}{\lfloor k/3 \rfloor}\right)n^2. \tag{1.1}$$

Dirac [2, p. 90] gave the $k = 6$ construction obtained by joining two odd cycles of equal order, and Erdős observed that it generalizes when $3 \mid k$, giving the coefficient in (1.1). When $3 \nmid k$, Toft's constructions [8] give larger coefficients. For background on dense critical graphs, see Jensen and Toft [6, Section 5.1, pp. 97–98] and Jensen [5]. Luo, Ma, and Yang [7] noted in 2023 that no constructions improving Toft's asymptotic constants had been found since 1970. We show that the coefficient in (1.1) is not optimal for $k = 12$.

Theorem 1. *There is a sequence of twelve-critical graphs G_s such that*

$$|V(G_s)| \longrightarrow \infty, \qquad \frac{e(G_s)}{|V(G_s)|^2} \longrightarrow \frac{2}{5}.$$

In particular, $f_{12}(n) \not\sim 3n^2/8$.

The graph G consists of five copies of Pegden's module $U(S,T)$ [9], with their active sets joined by an almost complete five-partite graph. The omitted edges form a graph Z whose adjacency is prescribed by a K_5-saturated graph H of small maximum degree from [1]. In an eleven-coloring each active set needs three colors (Lemma 2); the degree bound forces two of them to be private to that part. The eleventh color must then appear in all five parts, giving a K_5 in Z, impossible since H is K_5-free. Deleting an edge between two parts adds an edge to Z, and saturation supplies a K_5 through it, allowing one color to be shared in an eleven-coloring of $G - e$. Complete joins of critical graphs, as in Dirac's construction, cannot share colors between factors; omitting Z costs $o(n^2)$ edges and permits this shared color. Deleting the $o(n^2)$ module edges leaves a five-partite graph, so the asymptotic density of this construction is at most $2/5$. For comparison, the upper bound of Luo, Ma, and Yang [7, Theorem 1.1] has coefficient $9/20 - 1/4356 \approx 0.44977$ for f_{12}.

2 A coloring module

Let S be 11-critical and T be 10-critical. The module $U(S,T)$ consists of disjoint copies of S and T, together with an independent set

$$A = V(S) \times V(T).$$

Each $(x,y) \in A$ is adjacent to $x \in V(S)$ and $y \in V(T)$. There are no edges between S and T, and no other edges. We call A the active set and call $\{x\} \times V(T)$ the row indexed by x.

The following is the $U(11,10)$ case of Pegden's module lemma [9, Lemma 2.5], stated there for triangle-free factors; the proof below does not use that hypothesis.

Lemma 2. *Fix a palette of eleven colors.*

1. *In every proper coloring of $U(S,T)$ from this palette, the active set uses at least three colors.*
2. *Given $a_0 = (x_0, y_0) \in A$ and distinct colors α, β, γ, there is a proper coloring in which the active set uses only α, β, γ, and γ occurs on the active set only at a_0.*
3. *Given an edge e of $U(S,T)$ and distinct colors α, β, there is a proper coloring of $U(S,T) - e$ in which the active set uses only α, β.*

Proof. For (1), suppose the active colors are contained in $\{\alpha, \beta\}$. Since S uses all eleven colors, it contains a vertex colored α and a vertex colored β. Every active vertex in the first row must have color β, so no vertex of T can have color β. The second row similarly excludes α from T. This leaves only nine colors for T, a contradiction.

For (2), color S so that α occurs only at x_0. Color T with the ten colors other than α, so that β occurs only at y_0. Extend these colorings by

$$\operatorname{col}(x,y) = \begin{cases} \alpha, & x \neq x_0, \\ \beta, & x = x_0,\ y \neq y_0, \\ \gamma, & (x,y) = (x_0,y_0). \end{cases} \tag{2.1}$$

Each active vertex differs in color from both its neighbors.

For (3), there are four possibilities for the deleted edge. If $e \in E(S)$, color both $S - e$ and T with the ten colors other than α, and color all of A with α. If $e \in E(T)$, color $T - e$ with the nine colors other than α, β. Choose $x_0 \in V(S)$ and color S with α only at x_0. Color the row indexed by x_0 with β and all other rows with α.

Suppose instead that e joins an active vertex (x_0, y_0) to one of its two structural neighbors. Use the colorings of S and T from the proof of (2). If the deleted edge joins (x_0, y_0) to x_0, use (2.1) with the color of (x_0, y_0) changed to α. If it joins (x_0, y_0) to y_0, change that color to β. In either case the only edge that would become monochromatic is the deleted edge. $\square$

3 From saturated graphs to twelve-critical graphs

A graph is K_5-saturated if it contains no K_5, but adding any missing edge creates a K_5. Thus the common neighborhood of every pair of distinct nonadjacent vertices contains a triangle.

Proposition 3. *Let H be a K_5-saturated graph on $v \geq 5$ vertices, with maximum degree $d < v - 1$. Let $h \geq 11$ be odd and suppose*

$$v > 40hd. \tag{3.1}$$

Put $a = 2hv$. There is a twelve-critical graph G with

$$|V(G)| = 5(a + h + 2v) \tag{3.2}$$

and

$$e(G) \geq 10a^2 \left(1 - \frac{d}{v}\right). \tag{3.3}$$

Proof. Construction. Set $\ell = 2v$ and take the joins of a clique and an odd cycle in [2, p. 90]:

$$S = K_8 \vee C_{h-8}, \qquad T = K_7 \vee C_{\ell-7}.$$

Both cycles are odd and have length at least three. Since the join of a p-critical and a q-critical graph is $(p+q)$-critical, S is 11-critical and T is 10-critical. (Deleting a vertex or an edge within one factor saves a color there; if a joining edge xy is deleted, give x and y singleton colors in their respective factors and identify those colors.)

Take five disjoint copies $U_1, \dots, U_5$ of $U(S,T)$, with active sets $A_1, \dots, A_5$, each of size $a = h\ell$. In each copy, identify $V(T)$ with $V(H) \times \{0, 1\}$ by any bijection. An active vertex then has the form $(x, (z, \varepsilon))$; give it label z. Let π denote this labeling map. Each label occurs exactly $b = 2h$ times in each active set.

Define a five-partite graph Z on $A_1 \cup \cdots \cup A_5$ by putting, for $u \in A_i$ and $w \in A_j$ with $i \neq j$,

$$uw \in E(Z) \iff \pi(u)\pi(w) \in E(H).$$

The graph G retains all the edges of the five modules and, between different active sets, takes the complement of Z. There are no other edges between modules. In particular, active vertices with equal labels in different parts are adjacent in G.

We record three properties of Z. First, Z is K_5-free: the labeling map sends every clique injectively to a clique of H. Moreover, adding any missing edge between different parts of Z creates a K_5 with one vertex in each part. To see this, let u, w be the endpoints of such a missing edge. If their labels z, z' differ, saturation of H gives a triangle in $N_H(z) \cap N_H(z')$. Choose vertices with these three labels in the other three parts. Together with u, w, they form the required clique in $Z + uw$. If the labels coincide, say both equal z, choose a vertex z' not adjacent to z; this is possible because $d < v - 1$. The same common-neighborhood triangle is contained in $N_H(z)$ and gives the required clique.

Second, Z contains a K_4 across any prescribed four parts. Indeed, H has a missing edge; one endpoint together with a triangle in the common neighborhood gives a K_4 in H. Its four labels can be placed in any four parts of Z.

Third, $D := \Delta(Z) = 4bd = 8hd$, so (3.1) gives $\ell > 10D$.

The chromatic lower bound. Suppose that G has an eleven-coloring. Fix a part A_i and any color α. Since its copy of S uses all eleven colors, some vertex x of S has color α. The ℓ active vertices in the row indexed by x avoid α, so the pigeonhole principle and $\ell > 10D$ give a color $\beta \neq \alpha$ occurring more than D times in that row. Applying this observation twice, we may choose two distinct colors α_i, β_i, each used more than D times in A_i.

A color used more than D times in A_i cannot occur in any other active set. A vertex of that color in another part would have to be adjacent in Z to all those more than D vertices, contrary to $\Delta(Z) = D$. Thus the ten chosen colors $\alpha_1, \beta_1, \ldots, \alpha_5, \beta_5$ are distinct, leaving a single color γ. The active set A_i can use only $\alpha_i, \beta_i, \gamma$, and Lemma 2(1) forces γ to occur in every part. Choosing a vertex of this color in each part gives a K_5 in Z, a contradiction. Thus $\chi(G) \geq 12$.

Colorings after edge deletion. Use an eleven-color palette

$$\alpha_1, \beta_1, \ldots, \alpha_5, \beta_5, \gamma.$$

The pair α_i, β_i will be used only in A_i among the active sets. Structural vertices may use the full palette, since they have no neighbors in other modules.

Let $e = uw$ join two different active sets in G. By the first property of Z, there is a K_5 in $Z + uw$ containing uw, with vertices $a_i \in A_i$. Apply Lemma 2(2) in each module, using active colors $\alpha_i, \beta_i, \gamma$ and assigning γ only to a_i in A_i. The only color shared by different active sets is γ. The five vertices of this color are pairwise nonadjacent in $G - e$, because they form a clique in $Z + uw$. We have obtained an eleven-coloring of $G - e$.

Now let e lie within U_i. Apply Lemma 2(3) to color $U_i - e$, using only α_i, β_i on A_i. Choose a K_4 in Z across the other four parts. In each of those four modules, apply Lemma 2(2) with the selected vertex as the only active vertex of color γ. Once again this is a proper eleven-coloring of $G - e$.

These cases cover every edge of G, so $G - e$ is eleven-colorable for every edge e; recoloring one endpoint of a deleted edge with a twelfth color gives $\chi(G) \leq 12$. Since G has no isolated vertices, every proper subgraph of G lies in some $G - e$. Hence G is twelve-critical.

Counting vertices and edges. Each module has $a + h + 2v$ vertices, giving (3.2). Between a fixed pair of active sets, Z has $2b^2m$ edges: each edge of H gives two ordered pairs of labels, with b^2 pairs of active vertices for each, where $m = e(H)$. Since $2b^2m \leq b^2vd$ and $a = bv$, at least $10a^2(1 - d/v)$ edges lie between active sets, proving (3.3). □

Remark (Exact edge count)*.* The construction in Proposition 3 satisfies

$$e(G) = 10(a^2 - 8h^2m) + 5(2a + 9h + 16v - 79), \qquad m = e(H). \tag{3.4}$$

Indeed, each pair of active sets contributes $a^2 - 2b^2m$ edges, while $e(S) = 9h - 44$, $e(T) = 16v - 35$, and each module has $2a$ edges incident with its active set.

4 Choice of the saturated graphs

We use the following instance of the construction in [1, Section 4, Example 2].

Lemma 4 (Alon–Erdős–Holzman–Krivelevich)**.** *For every prime power $Q \geq 4$, there is a K_5-saturated graph H_Q with*

$$|V(H_Q)| = 13Q^2 + 12Q, \qquad \Delta(H_Q) = 22Q - 3. \tag{4.1}$$

Proof. In Example 2 of [1], set $k = 5$ and give each of the Q^2 additional independent sets one vertex. The core has $12Q(Q+1)$ vertices. A core vertex has $3(3Q-1)$ neighbors in its level, $12Q$ in the other levels, and Q among the additional vertices, so its degree is $22Q - 3$. An additional vertex has degree $12(Q+1)$, which is smaller for $Q \geq 4$. Thus (4.1) holds; saturation is part of the cited construction. □

Proof of Theorem 1. For each integer $s \geq 4$, set

$$h = 3^s, \qquad Q = h^2,$$

and take $H = H_Q$ from Lemma 4. Write $v = |V(H)|$ and $d = \Delta(H)$. Then

$$v = 13h^4 + 12h^2, \qquad d = 22h^2 - 3 < v - 1.$$

The integer h is odd and at least 81. Moreover,

$$40hd < 880h^3 < 13h^4 \leq v,$$

since $13h \geq 13 \cdot 81 > 880$. Proposition 3 therefore gives a twelve-critical graph G_s. Put $a = 2hv$ and $N_s = |V(G_s)|$. We have

$$\frac{N_s}{5a} = 1 + \frac{1}{h} + \frac{1}{2v} \longrightarrow 1, \qquad \frac{d}{v} \longrightarrow 0. \tag{4.2}$$

There are at most $10a^2$ edges between active sets, and the five modules contain $5(2a+9h+16v-79) = O(a)$ edges. Together with (3.3), this gives

$$10a^2\left(1 - \frac{d}{v}\right) \leq e(G_s) \leq 10a^2 + O(a).$$

Dividing by N_s^2 and using (4.2) yields

$$\frac{e(G_s)}{N_s^2} \longrightarrow \frac{10}{25} = \frac{2}{5}.$$

Finally, $N_s \to \infty$ and $f_{12}(N_s) \geq e(G_s)$. Since $2/5 > 3/8$, this subsequence rules out $f_{12}(n) \sim 3n^2/8$. □

Remark (Other chromatic numbers)*.* For each fixed $t \geq 3$, the same construction uses t modules, with

$$S = K_{2t-2} \vee C_{h-2t+2}, \qquad T = K_{2t-3} \vee C_{2v-2t+3},$$

of chromatic numbers $2t+1$ and $2t$, and a K_t-saturated graph H of order v and maximum degree d. Take $h \geq 2t+1$ odd and $v \geq t$, with $d < v-1$ and $v > 2t(t-1)hd$. Then $D = 2h(t-1)d$ and $2v > 2tD$. In a $(2t+1)$-coloring, each part has two private colors, and the remaining color would give a K_t in Z. Saturation supplies the transversal K_t after a cross-part edge deletion, and a K_{t-1} in the other parts after an internal deletion. The module colorings therefore give a $(2t+2)$-critical graph. By [1, Theorem 7], we may take $v \to \infty$ with $d = O_t(\sqrt{v})$, then choose odd $h \to \infty$ with $h = o(\sqrt{v})$. Writing $a = 2hv$, the order is $t(a+h+2v) \sim ta$, the cross-part edge count is $\binom{t}{2}a^2(1-o(1))$, and the module edges contribute $O_t(a)$. Thus every even $k = 2t+2 \geq 8$ admits a sequence with density

$$\frac{t-1}{2t} = \frac{k-4}{2(k-2)}.$$

Joining these graphs to K_1 gives, for every odd $k \geq 9$, a k-critical sequence with density $(k-5)/(2(k-3))$.

Lean formalization

Lemma 2, Proposition 3 (criticality, order, and the exact edge count (3.4)), the parameter choices, and the limit $e(G_s)/|V(G_s)|^2 \to 2/5$ have been formalized in Lean 4 with Mathlib. The formalization also defines f_{12} under the edge-deletion convention and proves that $f_{12}(n)/n^2$ has no limit below 2/5. The only unformalized input to these results is the existence of the graphs H_Q of Lemma 4, stated as an explicit hypothesis. Source and build instructions: `https://github.com/FireflySentinel/erdos-917`.

Use of generative AI

The proofs and the first draft were generated with GPT-6 Astra; GPT-5.6 Sol and Claude Opus 5 were used for editorial review; the Lean formalization was developed with OpenAI Codex (GPT-6). The author checked the arguments and is responsible for the content.

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